

INFERENCE STATISTICS → results produced for population on basis of sample.

↳ population (μ)

↳ sample: part of population but true representative. ($\bar{x}$)

$\bar{x}$ is statistic or estimated.

$\bar{x}$ is sample mean, μ is population mean, $E(x)$ is random variable mean.

$$V(x) = E(x)^2 - (E(x))^2$$

SAMPLING DISTRIBUTION → probability distribution of an estimator.

Standard error is standard deviation of sampling distribution of an estimator. $SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$

calculating standard error

↳ make samples

↳ mean

↳ find SD.

Q consider elements of population 2, 4, 6, 8

a) all possible samples of size 2 with replacement

b) show that $\mu_{\bar{x}} = \mu$ $S = 5$

c) show that $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$ $2.5 = \frac{5}{2}$

a) possible samples = $N^n = 4^2 = 16$

b) mean of sampling distribution equals to mean of population

$$\mu_{\bar{x}} = \sum \bar{x} f(\bar{x})$$

$$\sigma_{\bar{x}}^2 = \sum \bar{x}^2 f(\bar{x}) - [\sum \bar{x} f(\bar{x})]^2$$

$$\mu = \frac{\sum x}{n}$$

$$\sigma^2 = \frac{\sum x^2}{N} - \left(\frac{\sum x}{N} \right)^2$$

Samples	Mean
2,2	2
2,4	3
2,6	4
2,8	5
4,2	3
4,4	4
4,6	5
4,8	6
6,2	4
6,4	5
6,6	6
6,8	7
8,2	5
8,4	6
8,6	7
8,8	8

n = sample size

N = population size.

sampling distribution table:

$\bar{x}$	Tally	f	$f(\bar{x})$	$\bar{x} f(\bar{x})$
2	I	1	1/16	2/16
3	II	2	2/16	6/16
4	III	3	3/16	12/16
5	IIII	4	4/16	20/16
6	III	3	3/16	18/16
7	II	2	2/16	14/16
8	I	1	1/16	8/16
				<u>80/16</u>

$\bar{x}^2 f(\bar{x})$

4/16

18/16

48/16

100/16

108/16

98/16

64/16

440/16

x x^2

2 4

4 16

6 36

8 64

20 120

$\mu_{\bar{x}} = 5$

$\sigma_{\bar{x}}^2 = \frac{440}{16} - 5^2$

$\mu = 5$

$\sigma^2 = \frac{120}{4} - 5^2$

Q Population 3, 6, 9

a) sample size 3 with replacement

b) $\mu_{\bar{x}} = \mu$

c) $\sigma^2_{\bar{x}} = \frac{\sigma^2}{n}$

Q Population 3, 6, 9, 12, 15

a) sample size 2 without replacement

b) $\mu_{\bar{x}} = \mu$ $\mu = 9$

$\mu_{\bar{x}} = 90/10 = 9$

$\sigma^2_{\bar{x}} = \frac{877.5}{10} - (9)^2 = 6.75$

$\mu = \frac{3+6+9+12+15}{5} = 9$

$\sigma^2 = \frac{495}{5} - \left(\frac{45}{5}\right)^2 = 18$

c) $\sigma^2_{\bar{x}} = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right) = 6.75 = 9 \left(\frac{5-2}{5-1} \right) = 6.75$

a) $NC_n = SC_2 = 10$

Samples	mean	$\bar{x}$	Tally	f	f(x)	$\bar{x}f(\bar{x})$	$\bar{x}^2$	$\bar{x}^2f(\bar{x})$
3, 6	4.5	4.5		1	4.5/10	4.5/10	20.25	20.25/10
3, 9	6	6		1	6/10	6/10	36	36/10
3, 12	7.5	7.5		2	15/10	15/10	56.25	112.5/10
3, 15	9	9		2	18/10	18/10	81	162/10
6, 9	7.5	7.5		2	15/10	15/10	56.25	112.5/10
6, 12	9	9		2	18/10	18/10	81	162/10
6, 15	10.5	10.5		2	21/10	21/10	110.25	220.5/10
9, 12	10.5	10.5		2	21/10	21/10	110.25	220.5/10
9, 15	12	12		1	12/10	12/10	144	144/10
12, 15	13.5	13.5		1	13.5/10	13.5/10	182.25	182.25/10
				10	90/10	90/10		877.5/10

Q Population 6, 9, 15, 18

a) sample size 3 without replacement

Samples	mean	$\bar{x}$	Tally	f	f(x)	$\bar{x}f(\bar{x})$	$\bar{x}^2$	$\bar{x}^2f(\bar{x})$
6, 9, 15	10	10		3	30/10	30/10	100	300/10
6, 9, 18	11	11		2	22/10	22/10	121	242/10
6, 15, 18	13	13		1	13/10	13/10	169	169/10
9, 15, 18	14	14		1	14/10	14/10	196	196/10
				10	108/10	108/10		1206/10

$\mu_{\bar{x}} = 108/10 = 10.8$

$\mu = \frac{6+9+15+18}{4} = 10.8$

$\sigma^2_{\bar{x}} = \frac{1206}{10} - (10.8)^2 = 3.96$

$\sigma^2 = \frac{702}{5} - \left(\frac{54}{5}\right)^2 = 23.76$

$\mu_{\bar{x}} = \mu = 10.8 = 10.8$

$\sigma^2_{\bar{x}} = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right)$

$3.96 = \frac{23.76}{5} \left(\frac{5-3}{5-1} \right)$

Q Population 3,6,9

a) sample size 3 with replacement $N^n = 3^3 = 27$

b) $\mu_{\bar{x}} = \mu$

c) $\sigma^2_{\bar{x}} = \frac{\sigma^2}{n}$

Samples	Mean	$\bar{x}$	Tally	f	f($\bar{x}$)	$\bar{x}f(\bar{x})$	$\bar{x}^2$	$\bar{x}^2 f(\bar{x})$
3,3,3	3	3		1	1/27	3/27	9	9/27
3,3,6	4	4		3	3/27	12/27	16	48/27
3,3,9	5	5		6	6/27	30/27	25	150/27
3,6,3	4	6		7	7/27	42/27	36	252/27
3,6,6	5	7		6	6/27	42/27	49	294/27
3,6,9	6	8		3	3/27	24/27	64	192/27
3,9,3	5	9		1	1/27	9/27	81	81/27
3,9,6	6							
3,9,9	7			27		162/27		1026/27
6,3,3	4							
6,3,6	5							
6,3,9	6							
6,6,3	5							
6,6,6	6							
6,6,9	7							
6,9,3	6							
6,9,6	7							
6,9,9	8							
9,3,3	5							
9,3,6	6							
9,3,9	7							
9,6,3	6							
9,6,6	7							
9,6,9	8							
9,9,3	7							
9,9,6	8							
9,9,9	9							

$$\mu_{\bar{x}} = \frac{162}{27} = 6$$

$$\sigma^2_{\bar{x}} = \frac{1026}{27} - (6)^2 = 2$$

$$\mu = \frac{3+6+9}{3} = 6$$

$$\sigma^2 = \frac{126}{3} - (6)^2 = 6$$

$$\sum \bar{x}^2 = 3^2 + 6^2 + 9^2 = 126$$

$$\sum x = 6$$

$$\mu_{\bar{x}} = \mu \rightarrow 6 = 6$$

$$\sigma^2_{\bar{x}} = \frac{\sigma^2}{n} \rightarrow 2 = \frac{6}{3}$$

Estimator are statistics used for different random variables.

Random variable is a real value function whose domain is sample space.

Estimation is a process which we calculate estimators.

TYPES OF ESTIMATE

*Recap of SND and GND

- Point Estimate $\rightarrow$ If we have a single value it will be point estimate
- Interval Estimate. $\rightarrow$ If we have range of values for analysis.

$$x \sim N(\mu, \sigma^2) \rightarrow x \sim N(4, 9)$$

x follows normal distribution having mean 4 and variance 9

CENTRAL LIMIT THEORY

If $x_1, x_2, \dots, x_n$ be the random sample drawn from a particular population (any distribution), then

$$1. \sum_{i=1}^n x_i \sim N(n\mu, n\sigma^2)$$

$$2. \bar{x} \sim N(\mu, \frac{\sigma^2}{n})$$

PROBABILITY DISTRIBUTION

• DRV

$$\hookrightarrow E(x) = \sum x P(x)$$

$$\sum P(x) = 1$$

$$E(x^2) = \sum x^2 P(x)$$

• CRV

$$\hookrightarrow E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$\hookrightarrow \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\hookrightarrow \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$V(x) = E(x^2) - [E(x)]^2$$

UNIFORM DISTRIBUTION

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

$$E(x) = \frac{a+b}{2} \rightarrow E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$1. \int_{-\infty}^{\infty} f(x) dx = 1$$

$$2. E(x) = \frac{a+b}{2}$$

$$3. V(x) = \frac{(a-b)^2}{12}$$

$$= \int_a^b x \frac{1}{b-a} dx$$

$$= \frac{1}{b-a} \int_a^b x dx$$

$$= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b$$

$$= \frac{1}{b-a} \times \frac{(b-a)(b+a)}{2}$$

$$= \frac{a+b}{2}$$

$$\begin{aligned}
 v(x) &= \frac{(a-b)^2}{12} \rightarrow E(x^2) = \int_{-a}^b x^2 f(x) dx - \int x f(x) dx \\
 &= \int_a^b x^2 \frac{1}{b-a} dx - \left(\frac{a+b}{2} \right)^2 \\
 &= \frac{1}{b-a} \int_a^b x^2 dx - \left(\frac{a+b}{2} \right)^2 \\
 &= \frac{1}{b-a} \times \left| \frac{x^3}{3} \right| - \left(\frac{a+b}{2} \right)^2 \\
 &= \frac{1}{b-a} \times \frac{(b-a)(b^2+ab+a^2)}{3} - \frac{(a+b)^2}{4} \\
 &= \frac{b^2+ab+a^2}{3} - \frac{(a+b)^2}{4} \\
 &= \frac{b^2+ab+a^2 - (a+b)^2}{12} \\
 &= \frac{4a^2+4b^2+4ab - 3(a^2+b^2+2ab)}{12} \\
 &= \frac{4a^2+4b^2+4ab - 3a^2 - 3b^2 - 6ab}{12} \\
 &= \frac{a^2+b^2-2ab}{12}
 \end{aligned}$$

4. show that area under uniform curve is unity. $\int_a^b f(x) dx = 1$

$$\int_a^b \frac{1}{b-a} = 1$$

$$\frac{1}{b-a} \left| x \right|_a^b = 1$$

$$\frac{1}{b-a} (b-a) = 1$$

$$e^{ax+b} = \frac{e^{ax+b}}{a}$$

EXPONENTIAL DISTRIBUTION.

$$f(x) = \lambda e^{-\lambda x} \quad x > 0$$

$$1. E(x) = \frac{1}{\lambda}$$

$$2. V(x) = \frac{1}{\lambda^2}$$

3. area under exponential curve is unity.

$$\int_0^{\infty} \lambda e^{-\lambda x} = 1$$

$$\lambda \int_0^{\infty} e^{-\lambda x} = 1$$

$$\lambda \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^{\infty} = 1 \rightarrow (-1) [e^{-\infty} - e^{-0}] \rightarrow \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0 \rightarrow -1(0-1) = 1$$

$$2. E(x) = \int_0^{\infty} x \cdot \lambda e^{-\lambda x} dx = \lambda \int_0^{\infty} x e^{-\lambda x} dx$$

$$= \lambda \int uv - \int vu'$$

$$= \lambda \int_0^{\infty} \frac{-\lambda e^{-\lambda x}}{\lambda} + \left(\frac{1}{\lambda} \right) \left(\frac{e^{-\lambda x}}{-\lambda} \right)$$

$$= \lambda \left[0 + \left(\frac{1}{\lambda} \right) \left(\frac{e^{-\lambda x}}{-\lambda} \right) \right]_0^{\infty}$$

$$= \lambda \left(\frac{1}{\lambda^2} \right) = \lambda = \frac{1}{\lambda}$$

$$\begin{aligned} u &= x & v' &= e^{-\lambda x} \\ u' &= 1 & v &= \frac{e^{-\lambda x}}{-\lambda} \end{aligned}$$

$$(x) \left(\frac{e^{-\lambda x}}{-\lambda} \right) - \int \frac{e^{-\lambda x}}{-\lambda}$$

$$\frac{x e^{-\lambda x}}{-\lambda} - \left(\frac{-1}{\lambda} \right) \left(\frac{e^{-\lambda x}}{-\lambda} \right)$$

GAMMA DISTRIBUTION

$$f(x) = \frac{\lambda^p}{\Gamma(p)} x^{p-1} e^{-\lambda x}, \quad x > 0$$

$$\Gamma(p) = \int_0^{\infty} x^{p-1} e^{-x} dx.$$

when n is positive integer

$$\Gamma(n) = (n-1)!$$

1. show that area under gamma curve is 1

2. $E(x) = p/\lambda$

when n is fraction
 $\Gamma(n) = (n-1)\Gamma(n-1)$

3. $V(x) = p/\lambda^2$

$$\Gamma(\frac{1}{2}) = \sqrt{\pi}$$

$$\begin{aligned} \int_0^{\infty} \frac{\lambda^p}{\Gamma(p)} x^{p-1} e^{-\lambda x} dx &= \frac{\lambda^p}{\Gamma(p)} \int_0^{\infty} x^{p-1} e^{-\lambda x} dx \\ &= \frac{\lambda^p}{\Gamma(p)} \cdot \frac{\Gamma(p)}{\lambda^p} \\ &= 1 \end{aligned}$$

$$\int_0^{\infty} x^{n-1} e^{-ax} dx = \frac{\Gamma(n)}{a^n}$$

$$\begin{aligned} E(x) = p/\lambda &\rightarrow \int_0^{\infty} x \cdot \frac{\lambda^p}{\Gamma(p)} x^{p-1} e^{-\lambda x} dx \\ &= \frac{\lambda^p}{\Gamma(p)} \int_0^{\infty} x^p e^{-\lambda x} dx \\ &= \frac{\lambda^p}{\Gamma(p)} \cdot \frac{\Gamma(p+1)}{\lambda^{p+1}} \\ &= \frac{\lambda^p}{\Gamma(p)} \cdot \frac{\Gamma(p+1)}{\lambda^p \cdot \lambda} = \frac{p \Gamma(p)}{\Gamma(p) \times \lambda} = \frac{p}{\lambda} \end{aligned}$$

$$V(x) = \int_0^{\infty} x^2 f(x) dx - \int_0^{\infty} (x f(x))^2 dx$$

$$= \int_0^{\infty} x^2 \cdot \frac{\lambda^p}{\Gamma(p)} x^{p-1} e^{-\lambda x} dx - \left(\frac{p}{\lambda}\right)^2$$

$$= \frac{\lambda^p}{\Gamma(p)} \int_0^{\infty} x^{p+1} e^{-\lambda x} dx - \left(\frac{p}{\lambda}\right)^2$$

$$= \frac{\lambda^p}{\Gamma(p)} \cdot \frac{\Gamma(p+2)}{\lambda^{p+2}} - \left(\frac{p}{\lambda}\right)^2$$

$$p+1 = n-1$$

$$n = p+2$$

$$= \frac{\lambda^p}{\Gamma(p)} \cdot \frac{\Gamma(p+2)}{\lambda^p \cdot \lambda^2} - \frac{p^2}{\lambda^2}$$

$$= \frac{\lambda^p}{\Gamma(p)} \cdot \frac{(p+1)\Gamma(p+1)}{\lambda^p \cdot \lambda^2} - \frac{p^2}{\lambda^2}$$

$$\frac{(p+1) \cdot p \cancel{\lambda^p}}{\Gamma(p) \cdot \lambda^2} - \frac{p^2}{\lambda^2} = \frac{p^2 + p - p^2}{\lambda^2} = \frac{p}{\lambda^2}$$

EXPONENTIAL VARIANCE

$$V(x) = \int_0^{\infty} x^2 f(x) dx - \int_0^{\infty} [x f(x)]^2 dx$$

$$= \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx - \left(\frac{1}{\lambda}\right)^2$$

$$= \lambda \int_0^{\infty} x^2 e^{-\lambda x} dx - \frac{1}{\lambda^2}$$

$$u = x^2$$

$$v' = e^{-\lambda x}$$

$$u' = 2x$$

$$v = \frac{-e^{-\lambda x}}{\lambda}$$

$$= uv - \int v u' dx$$

$$= (x^2) \left(\frac{-e^{-\lambda x}}{\lambda} \right) - \int \frac{-e^{-\lambda x}}{\lambda} (2x) dx$$

$$= \frac{x^2 e^{-\lambda x}}{\lambda} - \left(\frac{-2}{\lambda} \right) \int x e^{-\lambda x} dx$$

$$= \lambda \left[\frac{-x^2 e^{-\lambda x}}{\lambda} - \left(\frac{-2}{\lambda} \right) \left[\frac{-x e^{-\lambda x}}{\lambda} + \frac{-e^{-\lambda x}}{\lambda^2} \right] \right]$$

$$= \frac{-x e^{-\lambda x}}{\lambda} + \left(\frac{1}{\lambda} \right) \left(\frac{e^{-\lambda x}}{-\lambda} \right)$$

$$= \lambda \left[\frac{-x^2 e^{-\lambda x}}{\lambda} \right] - \left[\frac{2x e^{-\lambda x}}{\lambda^2} + \frac{2e^{-\lambda x}}{\lambda^3} \right]_0^{\infty}$$

$$= -x^2 e^{-\lambda x} - \frac{2x e^{-\lambda x}}{\lambda} - \frac{2e^{-\lambda x}}{\lambda^2}$$

$$= \frac{+2}{\lambda^2} - \frac{1}{\lambda^2}$$

$$= \frac{1}{\lambda^2}$$

CENTRAL LIMIT THEORY

→ $X \sim$ any distribution

→ $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

Example 7.1

$$X \sim N(100, 10) \quad n = 25$$

$$P(\bar{X} < 95) = ?$$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{95 - 100}{10/\sqrt{25}}$$

$$Z = -2.5$$

$$P(Z < -2.5) = ?$$

$$\Phi(-2.5) = 0.0062$$

Example 7.2

$$f(x) = \begin{cases} \frac{1}{2} & 4 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases} \quad n = 40$$

$$\bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \rightarrow \mu = \frac{a+b}{2} = \frac{4+6}{2} = 5$$

$$\sigma^2 = \frac{(a-b)^2}{12} = \frac{(4-6)^2}{12} = \frac{1}{3}$$

$$\frac{\sigma^2}{n} = \frac{1}{3/40}$$

$$\bar{X} \sim N(5, \frac{1}{120})$$

* $\bar{X}$ follows normal distribution.

PROPERTIES OF ESTIMATORS.

BLUE $\rightarrow$ BEST LINEAR UNBIASED
ESTIMATOR.

1. Unbiasdness
2. Efficiency / Best
3. Consistency
4. Sufficiency.

1. UNBIASDNESS

- average value of estimator = true value of population
- an estimator $\hat{\theta}$ is said to be unbiased estimator of θ , if $E(x) \rightarrow E(\hat{\theta}) = \theta$
- $E(\hat{\theta}) - \theta = \text{bias}$

if both have bias, choose less

biasness and apply mean squared error.

Q Show that sample mean is an unbiased estimator of population mean.

$$E(\bar{x}) = \mu$$

Q Show that sample variance is unbiased estimator of population variance.

$$V(\bar{x}) \rightarrow E(s^2) = \sigma^2$$

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

$$\text{LHS} = E(s^2) = E\left(\frac{1}{n-1} \cdot \sum (x - \bar{x})^2\right)$$

$$\sum 1 = n$$

$$= \frac{1}{n-1} E\left(\sum ((x - \mu) + (\mu - \bar{x}))^2\right)$$

$$\sum 2 = 2 \sum 1 = 2n$$

$$= \frac{1}{n-1} E\left(\sum ((x - \mu)^2 + (\mu - \bar{x})^2 + 2(x - \mu)(\mu - \bar{x}))\right)$$

$$= \frac{1}{(n-1)} E\left[\sum (x - \mu)^2 + n(\bar{x} - \mu)^2 - 2(\bar{x} - \mu)\left(\sum (x - \mu)\right)\right]$$

$$= \frac{1}{n-1} E \left[\sum (x-\mu)^2 + n(\bar{x}-\mu)^2 - 2(\bar{x}-\mu) (\sum (x-\mu)) \right] \quad \sum (x-\mu)$$

$$= \frac{1}{n-1} E \left[\sum (x-\mu)^2 + n(\bar{x}-\mu)^2 - 2n(\bar{x}-\mu)^2 \right]$$

$$= \frac{1}{n-1} E \left[\sum (x-\mu)^2 - n(\bar{x}-\mu)^2 \right]$$

$$= \frac{1}{n-1} \left[\sum E(x-\mu)^2 - n E(\bar{x}-\mu)^2 \right]$$

$$= \frac{1}{n-1} \left[\sum V(x) - n V(\bar{x}) \right]$$

$$= \frac{1}{n-1} \left[\sum \sigma^2 - n \cdot \frac{\sigma^2}{n} \right]$$

$$= \frac{1}{n-1} \left[n\sigma^2 - \sigma^2 \right]$$

$$= \frac{1}{n-1} \cdot \sigma^2 (n-1)$$

$$= \sigma^2$$

$$V(x) = E(x^2) -$$

$$y = \frac{\sum (x - \bar{x})^2}{n} \quad a$$

$$= \sum \frac{a^2}{n} =$$

$$= E(x - \mu)$$

$$\frac{a^2}{n} \sum 1$$

$$\frac{a^2}{n} (n)$$

2. EFFICIENT

- If we have two estimators, then the estimator having min. variance will be best estimator.

Q We are given 2 estimator for population mean (μ)

$$\hat{\theta}_1 = \frac{x_1 + x_2 + x_3}{3}, \quad \hat{\theta}_2 = \frac{x_1 + 2x_2 + x_3}{4}$$

Which estimator is best?

1. check for unbiasedness of both estimator.

$$LHS = E(\hat{\theta}_1) = E\left(\frac{x_1 + x_2 + x_3}{3}\right)$$

$$= \frac{1}{3} E(x_1 + x_2 + x_3)$$

$$= \frac{1}{3} [E(x_1) + E(x_2) + E(x_3)]$$

$$= \frac{1}{3} [\mu + \mu + \mu] = \frac{1}{3} \cdot 3\mu = \mu$$

$$E[x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7]$$

$$\frac{1}{7} [\mu + \mu + \mu + \mu + \mu + \mu + \mu]$$

$$\frac{1}{7} [7\mu] = \mu$$

$$LHS = E(\hat{\theta}_2) = E\left(\frac{x_1 + 2x_2 + x_3}{4}\right)$$

$$= \frac{1}{4} E(x_1 + 2x_2 + x_3)$$

$$= \frac{1}{4} [E(x_1) + E(2x_2) + E(x_3)]$$

$$= \frac{1}{4} [\mu + 2\mu + \mu]$$

$$= \frac{1}{4} [4\mu] = \mu$$

$$\frac{1}{2} E[2\mu - \mu + \mu]$$

$$= \frac{1}{2} [2\mu] = \mu$$

$\therefore$ since both estimators are unbiased, check variance.

$$V(\hat{\theta}_1) = V\left(\frac{x_1 + x_2 + x_3}{3}\right)$$

$$= \frac{1}{9} V(x_1 + x_2 + x_3)$$

$$= \frac{1}{9} [V(x_1) + V(x_2) + V(x_3)]$$

$$= \frac{1}{9} [r^2 + r^2 + r^2] = \frac{1}{9} (3r^2) = \frac{1}{3} r^2$$

$$V(\hat{\theta}_2) = V(x_1 + 2x_2 + x_3)$$

4

$$= \frac{1}{16} V(x_1 + 2x_2 + x_3)$$

$$= \frac{1}{16} [V(x_1) + V(2x_2) + V(x_3)]$$

$$= \frac{1}{16} [\sigma^2 + 4\sigma^2 + \sigma^2]$$

$$= \frac{1}{16} [6\sigma^2] = \frac{3}{8} \sigma^2$$

$\therefore \hat{\theta}_1$ has less variance, hence best estimator.

METHOD OF MOMENTS

• let $x_1, x_2, \dots, x_n$ be a random sample of size 'n', then the r^{th} sample moment about zero is

$$\rightarrow m'_r = \frac{1}{n} \sum_{i=1}^n x_i^r \quad r=1,2,3, \dots$$

* parameters value cannot be changed

• the r^{th} moment of a random is given by

$$\rightarrow u'_r = E(x^r)$$

* variables value can be changed.

• equations formed depends on number of parameters

$$\rightarrow m'_r = u'_r$$

Q let $x_1, x_2, \dots, x_n$ be a random sample from poisson distribution with parameter λ . Find the method of moments estimator of λ

* estimator denoted

$$f(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \quad x=0,1,2,3, \dots$$

by λ

$\therefore$ since parameter is only one, so one equation is formed i.e. $u'_1 = m'_1 \rightarrow \textcircled{1}$

$\therefore$ poisson distribution $u'_1 = \lambda$ and $m'_1 = \frac{\sum x}{n} = \bar{x}$

$\therefore$ hence $m'_1 = u'_1 \rightarrow \bar{x} = \hat{\lambda}$

METHOD OF MAXIMUM LIKELIHOOD

Q Bernoulli Distribution $f(x; p) = f(x) = p^x q^{1-x}$, $x=0,1$ $0 \leq p \leq 1$, Find MLE

STEP 1 $L(p) = f(x_1; p) \cdot f(x_2; p) \dots f(x_n; p)$

$$= (p^{x_1} \cdot (1-p)^{1-x_1}) (p^{x_2} \cdot (1-p)^{1-x_2}) \dots (p^{x_n} \cdot (1-p)^{1-x_n})$$

$$L(p) = p^{\sum x_i} \cdot (1-p)^{n-\sum x_i}$$

single (exponential, poisson, bernoulli, etc)

STEP 2 Taking ln on both sides

$$l = \ln L(p) = \ln(p^{\sum x_i} \cdot (1-p)^{n-\sum x_i})$$

two (binomial, weibull)

$$l = \sum x_i \ln(p) + (n - \sum x_i) \ln(1-p)$$

STEP 3 now put $\frac{\partial l}{\partial p} = 0$

* derivative with respect to parameter

$$\sum x_i \cdot \frac{1}{p} \cdot 1 + (n - \sum x_i) \ln(-1) = 0$$

$$1-p$$

$$\frac{\sum x_i}{p} = (n - \sum x_i) / (1-p) \rightarrow \frac{\sum x_i}{p} = \frac{n - \sum x_i}{1-p}$$

$$\sum x_i - p \sum x_i = np - p \sum x_i$$

$$\sum x_i = np \rightarrow p = \frac{\sum x_i}{n} \rightarrow p = \bar{x} \rightarrow \hat{p} = \bar{x}$$

Q Find MLE of mean (μ) and variance (σ^2) of $N(\mu, \sigma^2)$

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2} \quad -\infty < x < \infty$$

$$L(\mu, \sigma^2) = \left(\frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \sigma^{-2} (x_1 - \mu)^2} \right) \left(\frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \sigma^{-2} (x_2 - \mu)^2} \right) \dots \left(\frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \sigma^{-2} (x_n - \mu)^2} \right)$$

$$= \left(\frac{1}{\sigma \sqrt{2\pi}} \right)^n e^{-\frac{1}{2} \sigma^{-2} [(x_1 - \mu)^2 + (x_2 - \mu)^2 + \dots + (x_n - \mu)^2]}$$

$$= \left(\frac{1}{\sigma \sqrt{2\pi}} \right)^n e^{-\frac{1}{2} \sigma^{-2} \sum_{i=1}^n (x_i - \mu)^2}$$

$$L(\mu, \sigma^2) = \left(\frac{1}{\sigma^2 2\pi} \right)^{n/2} e^{-\frac{1}{2} \sigma^{-2} \sum (x_i - \mu)^2}$$

$$\ln L(\mu, \sigma^2) = 0 - \frac{n}{2} \ln \sigma^2 - \frac{n}{2} \ln(2\pi) + \left(-\frac{1}{2} \sigma^{-2} \right) \sum_{i=1}^n (x_i - \mu)^2$$

$$\text{now } \frac{\partial l}{\partial \mu} = 0 \rightarrow -\frac{1}{2} \sigma^{-2} \sum 2(x_i - \mu) \cdot (-1) \rightarrow \frac{1}{\sigma^2} \sum (x_i - \mu) = 0 \rightarrow \sum (x_i - \mu) = 0 \rightarrow$$

$$\sum x_i - \sum \mu = 0 \rightarrow \sum x_i - n\mu = 0 \rightarrow \sum x_i = n\mu \rightarrow \hat{\mu} = \bar{x}$$

$$\begin{aligned}
 \text{now } \frac{\partial l}{\partial \sigma^2} &= 0 \rightarrow -\frac{n}{2} \left(\frac{1}{\sigma^2} \right) - 0 + \frac{1}{2} (\sigma^2)^{-2} \sum (x - \mu)^2 = 0 & y = \frac{1}{\sigma^2} = (\sigma^2)^{-1} \\
 &\rightarrow -\frac{n}{2} \left(\frac{1}{\sigma^2} \right) = -\frac{1}{2} \left(\frac{1}{\sigma^4} \right) \sum (x - \mu)^2 & = -1(\sigma^2)^{-1} \\
 &\rightarrow \frac{n}{\sigma^2} = \frac{\sum (x - \mu)^2}{\sigma^4} & = \frac{-1}{\sigma^4} \\
 &\rightarrow n = \frac{\sum (x - \mu)^2}{\sigma^2} \\
 &\rightarrow \sigma^2 = \frac{\sum (x - \mu)^2}{n} \\
 &\rightarrow \hat{\sigma}^2 = s^2
 \end{aligned}$$

Find estimators of parameter of Poisson distribution using method of moments

Q Find estimators of exponential using MLE

$$f(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \rightarrow f(x) = \lambda e^{-\lambda x}$$

$$L = (\lambda e^{-\lambda x_1}) (\lambda e^{-\lambda x_2}) \dots (\lambda e^{-\lambda x_n})$$

$$L = \lambda^n e^{-\lambda \sum x_i}$$

$$l = \ln(L) = \ln(\lambda^n e^{-\lambda \sum x_i})$$

$$= n \ln(\lambda) - \lambda \sum x_i \ln e$$

$$= n \ln \lambda - \lambda \sum x_i$$

$$\frac{\partial l}{\partial \lambda} = n \cdot \left(\frac{1}{\lambda} \right) - \sum x_i = 0 \rightarrow \frac{n}{\lambda} = \sum x_i \rightarrow \frac{1}{\lambda} = \bar{x} \rightarrow \hat{\lambda} = \frac{1}{\bar{x}}$$

MEAN SQUARED ERROR

• use MSE in case of variance and biased.

$$MSE(\hat{\theta}) = E(\hat{\theta} - \theta)^2$$

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{Bias})^2$$

— if unbiased, then bias = 0

$$\text{hence } MSE = \text{var} + 0$$

• in case of unbiasedness, you may use variance or MSE

• less MSE, is the best estimator. / whether biased or unbiased.

CHAPTER 8 : STATISTICAL INTERVALS

$\bar{x} = 3.1 \rightarrow$ point estimate

$\therefore$ also known as **point estimate**

2.5 - 3.6 $\rightarrow$ interval estimate

Confidence Interval

α = level of significance, chances of committing error

α = confidence level, range

Point and Interval Estimate

for PE, we have 1 value ; for IE, we have range

Level of Significance

chances of committing an error

Confidence Level

chances of not committing an error.

(Z)

(t)

(α)

(α, n)

* if σ known

$L >$ no. of observations

* σ unknown,
 $n \geq 30$

* σ is unknown
 $n < 30$

C 1 : CONFIDENCE INTERVAL FOR MEAN (population = μ)

e 1 $\rightarrow \sigma$ is known

the $(1-\alpha) 100\%$ CI for μ .

$$\bar{x} - \frac{Z_{\alpha}}{2} \left(\frac{\sigma}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + \frac{Z_{\alpha}}{2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

LB UB

$\frac{Z_{\alpha}}{2} \rightarrow$ critical values

$\sigma \rightarrow$ pop SD.

$n \rightarrow$ no. of observations

e 2 $\rightarrow \sigma$ is unknown, $[n \geq 30]$

the $(1-\alpha) 100\%$ CI for ' μ ' is.

$$\bar{x} - \frac{Z_{\alpha}}{2} \left(\frac{s}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + \frac{Z_{\alpha}}{2} \left(\frac{s}{\sqrt{n}} \right)$$

$s =$ SD of sample.

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

case 3 $\rightarrow \sigma$ is unknown, $n < 30$

The $(1-\alpha) 100\%$ for CI is

$$\bar{x} - t_{\frac{\alpha}{2}(n-1)} \left(\frac{s}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + t_{\frac{\alpha}{2}(n-1)} \left(\frac{s}{\sqrt{n}} \right)$$

Q For a population $\sigma = 5$, $\bar{x} = 31$, $n = 40$. find 95% CI for μ

since σ is known, we use Z table

the $(1-\alpha) 100\%$ for CI μ is;

$$= \bar{x} - \frac{Z_{\alpha}}{2} \left(\frac{\sigma}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + \frac{Z_{\alpha}}{2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$= 31 - (1.96) \left(\frac{5}{\sqrt{40}} \right) \leq \mu \leq 31 + (1.96) \left(\frac{5}{\sqrt{40}} \right)$$

$$= 29.46 \leq \mu \leq 32.55$$

$\therefore$ We're 95% confident, that population mean lies between 29.46 and 32.55

Q A sample is taken for a population $\rightarrow 2, 3, 5, 6, 9$, Find 99% CI

since σ is unknown and $n < 30$, we use t table

$$\bar{x} = \frac{2+3+5+6+9}{5} = 5$$

$$s = \sqrt{\frac{30}{4}}$$

$$s = 2.73$$

$$\alpha = 0.01$$

$$\frac{\alpha}{2} = \frac{0.01}{2} = 0.005$$

$$\bar{x} - t_{\frac{\alpha}{2}(n-1)} \left(\frac{s}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + t_{\frac{\alpha}{2}(n-1)} \left(\frac{s}{\sqrt{n}} \right)$$

$$5 - (4.604) \left(\frac{2.73}{\sqrt{5}} \right) \leq \mu \leq 5 + (4.604) \left(\frac{2.73}{\sqrt{5}} \right)$$

RECAP OF INVERSE NORMAL DISTRIBUTION

one sided $\rightarrow Z_{\alpha}$

* two sided $\rightarrow Z_{\alpha/2}$

Q $\alpha = 0.05$ $\frac{\alpha}{2} = 0.025$

$\alpha = 0.05$, $Z_{\alpha} = ?$

now $P(Z < a) = 0.025$

$\Phi(a) = 0.025$

$a = \Phi^{-1}(0.025)$

$a = 1.96$

now $P(Z < a) = 0.05$

$a = 1.645$

Q If $Z_{\alpha/2} = 1.96$, $1 - \alpha = ?$

STEP 1 $P(Z < 1.96) = \Phi(a) = \Phi(1.96) = 0.9750$

STEP 2 confidence level = $2 \times (\text{answer from step 1 i.e. } 0.9750)$

$= 2 \times (0.9750 - 0.5)$

$1 - \alpha = 0.95 = 95\%$

CONFIDENCE INTERVAL FOR PROPORTION

$P = \frac{x}{n}$

CHAPTER 09: STATISTICAL INFERENCE

any statement about population parameter is called hypothesis

1st APPROACH H_0 (null hypothesis) } opposite $H_0 : (=, \geq, \leq)$
 H_1 (alternative hypothesis) } $H_1 : (\neq, <, >)$

1. H_0 and H_1

$H_0 : \mu = 60$
 $H_1 : \mu \neq 60$ } 2-tailed

$H_0 : \mu \geq 60$
 $H_1 : \mu < 60$ } one-tailed

$H_0 : \mu \leq 60$
 $H_1 : \mu > 60$ } one-tailed

2 level of significance $\rightarrow \alpha = 5\%, 1\%, 10\%$ etc. $\rightarrow$ given

3 Test Statistic $\rightarrow z, t, \chi^2$ etc

$z_c =$

$t_c =$

$\chi^2_c =$

4 Critical Region (CR) $\rightarrow$ use H_1

$H_1 (\neq)$ CR $|z| > \frac{z_\alpha}{2}$

$H_1 (>)$ CR $z_c > z_\alpha$

$H_1 (<)$ CR $z_c < -z_\alpha$

$H_0 : \mu = 50$ $H_1 : \mu \neq 50$

$z = -1.53$ $\alpha = 0.05$

$\frac{z_\alpha}{2} = 1.96$

CR $|z| > \frac{z_\alpha}{2}$

5 Conclusion

$| -1.53 | > 1.96 \rightarrow$ not to

If calculated value lies in the CR, reject H_0 otherwise do not reject

2ND APPROACH - CONFIDENCE INTERVAL

1. calculate confidence interval. If calculated value of test statistic lies in CI, do not reject H_0 ,
otherwise reject H_0

3RD APPROACH - P-VALUE

* p-value is probability value approach

* If p-value is less than α , reject H_0 , otherwise do not reject.

* p-value means exact value of α , for which we reject H_0

$$\left. \begin{array}{l} p = 0.00024 \\ \alpha = 0.05 \end{array} \right\} \text{since p-value less than } \alpha, \text{ reject } H_0$$

EXAMPLE : FOR A DATASET $n=30$, $\bar{x}=15.2$, $\alpha=5\%$, $\sigma=3$

CHECK CLAIM $H_0: \mu=15.8$

1. $H_0: \mu=15.8$

$H_1: \mu \neq 15.8$

2. $\alpha = 0.05$

3. since σ is known, so we shall use Z-test.

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \rightarrow Z = \frac{15.2 - 15.8}{3/\sqrt{30}} = -1.095$$

4. CR $|Z| > Z_{\alpha/2}$

$|Z| > 1.96$

5. since Z calculated does not lie in CR, do not reject.

CI $\rightarrow$ The $(1-\alpha)$ 100% for mean $\bar{x} - \frac{Z_{\alpha}}{2} \left(\frac{\sigma}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + \frac{Z_{\alpha}}{2} \left(\frac{\sigma}{\sqrt{n}} \right)$

CI (14.13, 16.27), since value of μ lies in CI, so do not reject H_0

we want to approach parameters using estimators, and we always want to use large sample size.

→ limitations

1. time 2. budget 3. other constraints.

LAW OF LARGE NUMBERS

- relation b/w parameters and estimators.
- as we increase sample size, estimator will approach to parameter.

as $n \rightarrow \infty$, estimator $\rightarrow$ parameter

case 1 → sample size when population mean is given.

$$n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{E} \right)^2 \quad \therefore E = \text{margin of error}$$

case 2 → sample size when population proportion is given.

$$n = \left(\frac{Z_{\alpha/2}}{E} \right)^2 \cdot (p) \cdot (q)$$

∴ by calculating sample size, we use population parameter.

Type I - Error

→ rejecting true null (RT)

Type II - Error

→ accepting false null (AF)

1. TESTING HYPOTHESIS ABOUT POPULATION MEAN

CASE 1 $\rightarrow$ σ is known $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

CASE 2 $\rightarrow$ σ is unknown, $n \geq 30$ $Z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

CASE 3 $\rightarrow$ σ is unknown, $n < 30$ $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

($\neq$) CR $|z| \geq Z_{\alpha/2}$

($<$) CR $z < -Z_{\alpha}$

($>$) CR $z > Z_{\alpha}$

($\neq$) CR $|t| \geq t_{\frac{\alpha}{2}}(n-1)$

($<$) CR $t < -t_{\alpha}(n-1)$

($>$) CR $t > t_{\alpha}(n-1)$

EXAMPLE $n=10, \bar{x}=27, \sigma=1.2, \alpha=1\%$

$H_0: \mu \leq 26.3$

$H_1: \mu < 26.3$

1. $H_0: \mu \leq 26.3$

$H_1: \mu > 26.3$

2. $\alpha = 0.01$

3. $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \rightarrow \frac{27 - 26.3}{1.2/\sqrt{10}} = 1.845$

4. $Z > Z_{\alpha}$

$Z > Z_{0.01}$

$Z > 2.33$

$\therefore$ since value does not lie in CR, so do not reject.

EXAMPLE $n = 49$, $\bar{x} = 125$, $\sigma = 4.2$

$$H_0: \mu > 123.5$$

$$H_1: \mu < 123.5$$

a) check H_0

$$1. H_0: \mu > 123.5$$

$$4. Z < -Z_{\alpha}$$

$$H_1: \mu < 123.5$$

$$Z < -Z_{0.10}$$

$$2. \alpha = 0.10$$

$$Z < -1.28$$

$$3. Z = \frac{125 - 123.5}{4.2/\sqrt{49}} = 2.5$$

5. do not reject

b) CI

$$(123.5, 126.5)$$

EXAMPLE Mean lifetime of lightbulbs is 1120 hours, with SD = 125 hours.

A sample of 100 electric bulbs showed mean lifetime of 1070 hours.

Company claimed that bulbs lifetime has not changed.

a) 5% level of significance

b) verify using CI.

$$a) n = 100, \bar{x} = 1070, \sigma = 125, \alpha = 0.05$$

$$H_0: \mu = 1070$$

$$H_1: \mu \neq 1070$$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{1070 - 1070}{125/\sqrt{100}} = -4$$

$$CR \quad |Z| \geq \frac{Z_{\alpha}}{2} = Z_{0.025} = 1.96$$

$4 > 1.96 \therefore$ calculated value lies in CR, reject H_0 .

b) CI

We always decide p-value on basis of H_1

$$H_1 (\neq) \quad P = 2[1 - \Phi(|z_0|)]$$

$$H_1 (>) \quad P = 1 - \Phi(z_0)$$

$$H_1 (<) \quad P = \Phi(z_0)$$

A sample of 36 observation selected from normal population

The sample mean is 21, SD is 5 of population. Using 5% level of significance, check claim that population mean > 20 .

a) critical value b) confidence interval c) p-value.

1. $n = 36, \sigma = 5, \bar{x} = 21, \alpha = 0.05$

2. $H_0: \mu \leq 20$

$H_1: \mu > 20$

3. $\alpha = 0.05$

4. $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{21 - 20}{5/6} = 1.2$

5. CR $z > z_\alpha$

$z > 1.65$

6. since calculated value of z does not lie in CR, so donot reject H_0 .

c) $H_0: \mu \leq 20$

$H_1: \mu > 20$

$\alpha = 0.05$

$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{21 - 20}{5/6} = 1.2$

$H_1 (>) \quad P = 1 - \Phi(z_0)$

$P = 1 - \Phi(1.2)$

$P = 1 - 0.8849$

$P = 0.1151$

$\therefore$ p-value is greater than α , donot reject.

$\therefore$ check $H_1 (>)$

Testing hypothesis about mean when σ is unknown

• 3 approaches.

1. Critical value
2. Confidence Interval
3. p-value

• CR $H_1: \mu \neq \mu_0$ $|t| > t_{\frac{\alpha}{2}}(n-1)$ $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

$H_1: \mu < \mu_0$ $t > t_{\alpha}(n-1)$

$H_1: \mu > \mu_0$ $t < -t_{\alpha}(n-1)$

EXAMPLE $n=6$, $\sum x = 9034.8$, $\sum (x - \bar{x})^2 = 50.8$ $\alpha = 10\%$

σ = population
standard
deviation.

CR $H_0: \mu = 1503$

$H_1: \mu \neq 1503$

a) CVA b) CIA

$$\bar{x} = \frac{\sum x}{n} \quad s = \frac{\sum (x - \bar{x})^2}{n-1}$$

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{1505.8 - 1503}{\sqrt{10.16}/\sqrt{6}}$$

$$\bar{x} = \frac{9034.8}{6} \quad s = \frac{50.8}{5}$$

$$t = 2.15$$

CR $|t| > 2.015$

$$\bar{x} = 1505.8 \quad s = \sqrt{10.16} = 3.18$$

$$\bar{x} - t_{\frac{\alpha}{2}}(n-1) \left(\frac{s}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + t_{\frac{\alpha}{2}}(n-1) \left(\frac{s}{\sqrt{n}} \right)$$

$$1505.8 - t_{(0.05)(5)} \left(\frac{\sqrt{10.16}}{\sqrt{6}} \right) \leq \mu \leq 1505.8 + t_{(0.05)(5)} \left(\frac{\sqrt{10.16}}{\sqrt{6}} \right)$$

$$1505.8 - (2.015) \left(\frac{\sqrt{10.16}}{\sqrt{6}} \right) \leq \mu \leq 1505.8 + (2.015) \left(\frac{\sqrt{10.16}}{\sqrt{6}} \right)$$

$$1497.44 \leq \mu \leq 1514.15$$

$$1503.17 \leq \mu \leq 1508.42$$

HYPOTHESIS TESTING ABOUT VARIANCE

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

CR $H_1 (\neq) : \chi^2 < \chi^2_{(1-\frac{\alpha}{2})(n-1)} , \chi^2 > \chi^2_{\frac{\alpha}{2}(n-1)}$

$H_1 (>) : \chi^2 > \chi^2_{\alpha(n-1)}$

$H_1 (<) : \chi^2 < \chi^2_{(1-\alpha)(n-1)}$

CI $\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2}(n-1)}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{(1-\frac{\alpha}{2})(n-1)}}$

EXAMPLE If $n=20, s^2 = 0.0153, \alpha = 0.05$

$H_0 : \sigma^2 \leq 0.01$

$H_1 : \sigma^2 > 0.01$

a) CVA b) CI

$$\chi^2 = \frac{(20-1)(0.0153)}{0.01} = 29.07$$

$H_1 (>) : \chi^2 < \chi^2_{(1-\alpha)(n-1)}$

$$\chi^2 < \chi^2_{(0.95)(19)}$$

$$\chi^2 < 10.117 \rightarrow \text{reject } H_0$$

$$\frac{(20-1)(0.0153)}{\chi^2_{(0.025)(19)}} \leq \sigma^2 \leq \frac{(20-1)(0.0153)}{\chi^2_{(0.975)(19)}}$$

$$\chi^2_{(0.025)(19)}$$

$$\chi^2_{(0.975)(19)}$$

$$\frac{(20-1)(0.0153)}{32.852} \leq \sigma^2 \leq \frac{(20-1)(0.0153)}{8.907}$$

$$32.852$$

$$8.907$$

$$0.0088 \leq \sigma^2 \leq 0.03263$$

Inference for single population.

1. mean $\begin{cases} \sigma \text{ known (z)} \\ \sigma \text{ unknown (t)} \end{cases}$

2. variance : $\begin{cases} H_0: \sigma^2 = 4 \\ \sigma \neq 4 \end{cases} \quad \chi^2 \rightarrow \frac{(n-1)s^2}{\sigma^2}, n-1$

3. proportion

TESTING HYPOTHESIS FOR PROPORTION

1. CV

2. H_0, H_1

3. α

4. Test statistic

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

1. CI

$$\hat{p} - Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}} < p < \hat{p} + Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

1. p-value approach

5. CR remains same

6. conclusion.

. in

In section c of DS, sample of 5 students selected at random. The CGPA is greater than 3.8. Suppose there are 2 students in sample of 5 having CGPA > 3.8. Check claim that proportion of students having CGPA > 3.8 is greater than 0.4, where $\alpha = 1\%$.

→ use all 3 approaches.

1. $H_0: P \leq 0.4$

$H_1: P > 0.4$

$$\hat{p} = \frac{x}{n} = \frac{2}{5} = 0.4$$

2. $\alpha = 0.01$

3. $Z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}} \rightarrow Z = \frac{0.4 - 0.4}{\sqrt{\frac{(0.4)(0.4)}{5}}} = 0 \rightarrow 0.5$

4. CR $Z_c > Z_\alpha$

$Z_c > Z_{0.01}$

$Z_c > 2.33$

5. calculated value does not lie in CR so do not reject.

CI $0.4 - (2.58) \sqrt{\frac{(0.4)(0.6)}{5}} < P < 0.4 + (2.58) \sqrt{\frac{(0.4)(0.6)}{5}}$
 $-0.165 < P < 0.965$

• calculated value lies so do not reject.

P-value.

$P = 1 - \Phi[Z_0]$

$P = 1 - \Phi[0]$

$P = 1 - 0.5$

$P = 0.5$

• calculated value is greater so do not reject.

Tests

Parametric

Non-parametric

follows a
specific
distribution

Non-parametric $\rightarrow$ observed

expected: calculated from model

NON-PARAMETRIC TEST - SECTION 9.7

1. GOODNESS OF FIT TEST

1. H_0 : good model

$$E = np$$

H_1 : not a good model (not H_0)

$$E = np$$

2. α = given or 0.05

3. Test statistic $\rightarrow \chi^2 = \sum \left(\frac{O-E}{E} \right)^2$

degree of freedom = $k-1$ $\therefore k$ = categories

4. CR $\chi^2_c > \chi^2_{\alpha}(df)$

5. If calculated values lie in CR so reject H_0 , otherwise do not reject H_0

CONTINGENCY TABLE - TWO WAY CLASSIFICATION

Provinces	FS-C Marks	50-60	60-70	70-80	80-90
P		A	C		
S					
B					
K					

Linear $\rightarrow x$ and y must be all

$$y = B_0 + B_1x_1 + B_2x_2 + \dots + B_kx_k \quad \text{variable} = k \quad \text{parameters} = k+1$$

$$y = B_0 + B_1x + \epsilon \quad \rightarrow \text{model so population parameter}$$

$\downarrow$ $\downarrow$
Sales adv. budget

$$y = 1000 + 3x$$

$\downarrow$ $\downarrow$
 $\hat{B}_0$ $\hat{B}_1$

$\rightarrow$ estimators, calculated from sample.

$\rightarrow$ even if adv. budget is 0, store sales will be 0

$\rightarrow$ adv. budget increased by 1, store sales increased by 3 times

$$H_0: \beta_1 = 0$$

$$H_1: \beta_1 \neq 0$$

TESTS OF STATISTICAL HYPOTHESES.

Type I Error $\rightarrow$ rejecting the null hypotheses H_0 , when it is true.

Type II Error $\rightarrow$ failing to reject null hypotheses when it is false

probability of Type I Error $\rightarrow \alpha = P(\text{type I error}) = P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$

probability of Type II Error $\rightarrow \beta = P(\text{type II error}) = P(\text{fail to reject } H_0, \text{ when } H_0 \text{ is false})$

Power = probability of rejecting null hypotheses H_0 , when H_1 is true.

$$1 - \beta$$

for one sided $\rightarrow$ rejecting H_0 is always a strong conclusion.

Reject H_0	Fail to reject H_0	Reject H_1
48.5	50	51.5

CHAPTER 10 • STATISTICAL INFERENCE USING TWO POPULATION/SAMPLES

TESTING HYPOTHESIS AND CONFIDENCE INTERVAL FOR DIFFERENCE OF 2 MEANS.

1. Critical Value Approach (σ_1, σ_2 known)

→ $H_0: \mu_1 - \mu_2 = \Delta$ (any value)

$H_1: \mu_1 - \mu_2 \neq \Delta$ (any value)

→ $\alpha = 1\%, 5\%, 10\%$ etc by default = 5%

→ test statistic

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

→ CR

($\neq$) : $|Z| \geq \frac{Z_\alpha}{2}$

(<) : $Z < -Z_\alpha$

(>) : $Z > Z_\alpha$

→ Conclusion

If lies in CR, then reject H_0

* estimator - table

standard error.

* mean estimator $\bar{x}$

parameter is μ

2. Confidence Interval

The $(1 - \alpha)$ 100% confidence interval of

$(\bar{x}_1 - \bar{x}_2) -$

$$(\bar{x}_1 - \bar{x}_2) - Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq (\bar{x}_1 - \bar{x}_2) + Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

. If value given in H_0 lies in CI, then don't reject H_0

→ lower bound for one-sided CI for ' $\mu_1 - \mu_2$ '

$$(\bar{x}_1 - \bar{x}_2) - Z_\alpha \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

3. P-value approach

→ H_0 :

H_1 :

→ $\alpha = 1\%, 5\%$

→ test statistic

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

→ CR

$$(\neq) : P = 2[1 - \Phi(|z_0|)]$$

$$(>) : P = 1 - \Phi(z)$$

$$(<) : P = \Phi(z_0)$$

→ If p-value less than α , then reject H_0

EXAMPLE : For two population , $\alpha = 0.05$, $H_0 : \mu_1 = \mu_2$ vs $H_1 : \mu_1 \neq \mu_2$

$$\sigma_1 = 10 \quad \sigma_2 = 5$$

a) CVA b) CI c) PVA

$$n_1 = 10 \quad n_2 = 15$$

$$\bar{x}_1 = 4.7 \quad \bar{x}_2 = 7.8$$

a) $H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$

$\alpha = 0.05$

$$Z = (4.7 - 7.8) - (0)$$

$$\sqrt{\frac{(10)^2}{10} + \frac{(5)^2}{15}}$$

$$Z = -0.90$$

CR ($\neq$) : $|Z| \geq \frac{Z_{\alpha}}{2}$

$| -0.90 | > 1.96$ ∴

does not lie in CR, so
do not reject

$$b) (4.7 - 7.8) - (1.96) \sqrt{\frac{(10)^2}{10} + \frac{(5)^2}{15}} \leq \mu_1 - \mu_2 \leq (4.7 - 7.8) + 1.96 \sqrt{\frac{(10)^2}{10} + \frac{(5)^2}{15}}$$

$$-9.794 \leq \mu_1 - \mu_2 \leq 3.594$$

value of H_0 lies in CI, so do not reject

c) $z = -0.90$

CR ($\neq$) : $P = 2[1 - \Phi(0.90)]$

$$P = 2[1 - 0.8159]$$

$$P = 0.3682$$

p-value greater than α , so do not reject H_0 .

$$(d) \quad (4.7 - 7.8) - (1.65) \sqrt{\frac{10^2}{10} + \frac{5^2}{15}} \leq \mu_1 - \mu_2$$

$$- 8.7 \leq \mu_1 - \mu_2$$

$$e) \quad (4.7 - 7.8) + (1.65) \sqrt{\frac{10^2}{10} + \frac{5^2}{15}}$$

$$\mu_1 - \mu_2 \leq 2.53$$

EXAMPLE 10.4

$$\sigma_1 = 0.020$$

$$\sigma_2 = 0.025$$

$$n = 10$$

$$n = 10$$

$$\bar{x} = 16.015$$

$$\bar{y} = 16.005$$

$$s^2 = 0.000916$$

$$s^2 = 0.00065$$

$$H_0 : \mu_1 = \mu_2$$

$$\mu_1 \neq \mu_2$$

$$\alpha = 0.05$$

$$Z = (16.015 - 16.005) -$$

$$CI = (16.015 - 16.005) - (1.96) \sqrt{\frac{(0.02)^2}{10} + \frac{(0.025)^2}{10}} \leq \mu_1 - \mu_2 \leq$$

$$(16.015 - 16.005) + (1.96) \sqrt{\frac{(0.02)^2}{10} + \frac{(0.025)^2}{10}}$$

$$- 0.0098 \leq \mu_1 - \mu_2 \leq 0.0298$$

WILCOXON SIGNED RANK TEST

$$1. \begin{matrix} H_0 : (\mu = \mu_0) & \left(\begin{matrix} \mu \leq \mu_0 \\ \mu > \mu_0 \end{matrix} \right) & \left(\begin{matrix} \mu > \mu_0 \\ \mu < \mu_0 \end{matrix} \right) \\ H_1 : (\mu \neq \mu_0) & & \end{matrix}$$

* parametric

follows a distribution

* non-parametric

does not follow of distribution

• it is a non-parametric test.

2. α = level of significance = 0.05, 0.01, etc.

3. Test statistic

$$T = \min(T^-, T^+)$$

T^- : sum of ranks of negative values in absolute terms

4. Critical Region

$$T_{cal} \leq T_{\alpha} \quad (\text{table value})$$

T^+ : sum of ranks of positive values.

5. Conclusion

If T lies in CR, then reject H_0

EXAMPLE : In a study of drug abused in a sub urban area, investigator found that

avg IB is 107. Researcher wishes to know whether avg IB is 107 for another sub-urban are. using data.

1. $H_0 : \mu = 107$

$H_1 : \mu \neq 107$

4. $T \leq T_{\alpha} \quad \alpha = 0.05, n = 14$

$T \leq 21 \quad \rightarrow 40.5 \leq 21$

2. $\alpha = 0.05$

5. Since T_{cal} does not lie in CR,

3. $T = \min(T^-, T^+)$

so do not reject H_0 .

$= \min(40.5, 64.5)$

$= 40.5$

Persons	IQ	$D_i = x_i - 107$	$ D_i $	Rank D_i	Rank D_i signed
1	99	-8	8	7	-7
2	100	-7	7	6	-6
3	90	-17	17	11	-11
4	94	-13	13	10	-10
5	135	28	28	14	14
6	108	1	1	1	1
7	107	0	0	-	-
8	111	4	4	5	5
9	119	12	12	9	9
10	104	-3	3	4	-4
11	127	20	20	13	13
12	109	2	2 ^{2nd}	2.5	2.5
13	117	10	10	8	8
14	105	-2	2 ^{3rd}	2.5	-2.5
15	125	18	18	12	12

$$T^- = 7 + 6 + 11 + 10 + 4 + 2.5 = 40.5$$

$$T^+ = 14 + 1 + 5 + 9 + 13 + 2.5 + 8 + 12 = 64.5$$

CHAPTER II - SIMPLE LINEAR REGRESSION AND CORRELATION

regression

variable

natural

controlled

RV

Fixed

dependent

* in regression, we study about natural variables on basis of controlled variables

$$y = f(x) \rightarrow \frac{dy}{dx}$$

$$Z_x = x^3 y^2 + 2x - 3y + 1$$

$$Z_x = 3x^2 y^2 + 2 - 0 + 0$$

$$Z_y = 2x^3 y + 0 - 3 + 0$$

SIMPLE LINEAR REGRESSION MODEL (SLRM)

a regression model in which one dependent, one independent variable, and parameter should be linear

$$y = \beta_0 + \beta_1 x + e$$

β_0 - y-intercept $\rightarrow \hat{\beta}_0$

β_1 - slope $\rightarrow \hat{\beta}_1$

e - error term

} estimators.

in order to drive estimator of SLRM, we use 3 methods;

1. Method of least squares.

2. MLE

3. Methods of Moments.

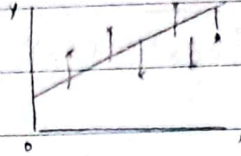
2. METHOD OF LEAST SQUARES.

$$\textcircled{1} S = \sum e = 0$$

$$e = y - \hat{y}$$

$$\textcircled{2} S = \sum |e| \quad (\text{Min})$$

$$\textcircled{3} S = \sum e^2$$



• we minimize sum of squares of errors.

PROOF-1 find estimator of SLRM

$$y = \beta_0 + \beta_1 x + e \quad \hat{\beta}_0 = ? \quad \hat{\beta}_1 = ? \quad \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$S = \sum e^2$$

$$S = \sum (y - \hat{y})^2$$

$$\frac{\partial S}{\partial \hat{\beta}_0} = \sum 2(y - \hat{\beta}_0 - \hat{\beta}_1 x)(-1)$$

$$\frac{\partial S}{\partial \hat{\beta}_1} = \sum 2(y - \hat{\beta}_0 - \hat{\beta}_1 x)(-x)$$

$$S = \sum (y - \hat{\beta}_0 - \hat{\beta}_1 x)^2$$

$$= \sum y - \sum \hat{\beta}_0 - \sum \hat{\beta}_1 x$$

$$\sum xy - \hat{\beta}_0 \sum x - \hat{\beta}_1 \sum x^2 = 0$$

$$\sum y = n\hat{\beta}_0 + \hat{\beta}_1 \sum x$$

$$\sum xy = \sum x\hat{\beta}_0 + \sum x^2 \hat{\beta}_1$$

$$\star y = \beta_0 + \beta_1 x$$

$$\sum y - \hat{\beta}_0 \sum 1 - \hat{\beta}_1 \sum x = 0$$

• check β_0 coefficient,
multiply and $\sum$

$$\sum y = n\hat{\beta}_0 + \sum x\hat{\beta}_1$$

$$\cdot \sum y = n\beta_0 + \beta_1 \sum x$$

$$\sum xy = \beta_0 \sum x + \beta_1 \sum x^2$$

$$\star y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

$$\star n\hat{\beta}_0 + \sum x\hat{\beta}_1 = \sum y$$

$$\sum y = n\beta_0 + \beta_1 \sum x_1 + \beta_2 \sum x_2$$

=

$$\sum xy = \beta_0 \sum x_1 + \beta_1 \sum x_1^2 + \beta_2 \sum x_1 x_2$$

$$\sum x_2 y = \beta_0 \sum x_2 + \beta_1 \sum x_1 x_2 + \beta_2 \sum x_2^2$$

after simplification we get $\hat{\beta}_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$

$$\frac{n \hat{\beta}_0}{n} + \frac{\sum x \hat{\beta}_0}{n} = \frac{\sum y}{n}$$

$$\hat{\beta}_0 + \frac{\sum x (\hat{\beta}_0)}{n} = \frac{\sum y}{n}$$

$$\hat{\beta}_0 = \hat{y} - \hat{\beta}_1 x'$$

PROOF-2 find estimator of SLRM using MLE

x	y	xy	x ²	y ²	(x - $\bar{x}$) ²
5	4	20	25	16	49
7	12	84	49	144	25
12	18	216	144	324	0
16	21	336	256	441	16
20	24	480	400	576	64

$$b) y = \beta_0 + \beta_1 x + e$$

$$\hat{\beta}_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = 1.22$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 1.16$$

$$\hat{y} = 1.16 + 1.22x$$

$$SSE = \sum e^2 = 23.29$$

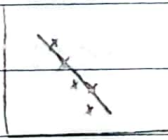
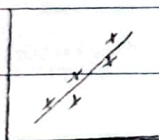
a) construct scatter plot

b) fit a reg. line $y = \beta_0 + \beta_1 x + e$

c) find SSE $S = \sum e^2$

3. SCATTER PLOT

• it tells us relation about two points whether positive or negative



4. CORRELATION COEFFICIENT

• it tells us about linear relationship of variables.

$$r = \frac{n \sum xy - \sum x \sum y}{[n \sum x^2 - (\sum x)^2] [n \sum y^2 - (\sum y)^2]}$$

$$[n \sum x^2 - (\sum x)^2] [n \sum y^2 - (\sum y)^2]$$

write x in place of y

$$r = 0.9528 \quad (\text{from the above example given})$$

• limitations of $r \rightarrow -1 \leq r \leq 1$

↳ if r is 1, it shows perfect positive correlation

↳ if r is -1, it shows perfect negative correlation.

↳ if r is 0.5, it shows moderate positive

↳ if r is -0.5, it shows moderate negative.

regression analysis

linear

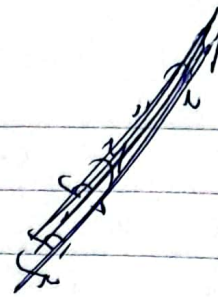
non-linear

SLRM

MLRM

$$y = \beta_0 + \beta_1 x + e$$

$$\hat{\beta}_0, \hat{\beta}_1$$



COEFFICIENT OF DETERMINATION (R^2)

tells us about goodness of fit of the model.

$R^2 = (r)^2$ ∴ square of correlation coefficient is equal to coefficient of determination.

$$R^2 = (0.9528)^2$$

$$R^2 = 0.9078 \approx 0.91$$

interpretation: 91% variation in y is due to x and remaining 9% is due to other factors.

limitations $0 \leq R^2 \leq 1$

if $R^2 = 1$, completely depends on x

close to 1, good model.

TESTING OF PARAMETERS OF THE MODEL.

Testing of β_1

① $H_0: \beta_1 = 0$ (importance is 0 i.e. insignificant)

$$H_1: \beta_1 \neq 0$$

② $\alpha = 0.05, 0.01, \text{etc}$

$$\textcircled{3} \quad t = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} = 5.43 \rightarrow SE(\hat{\beta}_1) = \sqrt{\frac{\sigma_e^2}{\sum (x - \bar{x})^2}} \rightarrow \sigma_e^2 = \frac{SE^2}{n-2}$$

$$\textcircled{4} \quad \text{CR } |t| \geq t_{\frac{\alpha}{2}, (n-2)} \quad SE(\hat{\beta}_1) = \sqrt{\frac{7.76}{154}} = 0.2245 \quad \sigma_e^2 = \frac{23.29}{5-2}$$

⑤ since calculated value lies in CR, so reject H_0 , hence

β_0 is significant

$$= 7.76$$

7. CI for $\hat{\beta}_1$:

$$\hat{\beta}_1 - t_{\alpha/2, n-2} SE(\hat{\beta}_1) \leq \beta_1 \leq \hat{\beta}_1 + t_{\alpha/2, n-2} (SE(\hat{\beta}_1))$$

$$1.22 - 3.18 (0.2245) \leq \beta_1 \leq 1.22 + 3.18 (0.2245)$$

$$0.50609 \leq \beta_1 \leq 1.93391$$

∵ since $\beta_1 = 0$, does not lie in CI, so reject H_0 , β_1 = significant, x = important.

8. MATRICES APPROACH

$$y = \beta_0 + \beta_1 x$$

$$2x + 3y = 5$$

$$\sum y = n\beta_0 + \beta_1 \sum x$$

$$5x + 4y = 9$$

$$\sum xy = \beta_0 \sum x + \beta_1 \sum x^2$$

$$\begin{pmatrix} n & \sum x \\ \sum x & \sum x^2 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} \sum y \\ \sum xy \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} n & \sum x \\ \sum x & \sum x^2 \end{pmatrix}^{-1} \begin{pmatrix} \sum y \\ \sum xy \end{pmatrix}$$

$$\frac{1}{n\sum x^2 - (\sum x)^2} \begin{pmatrix} n & \sum x \\ \sum x & \sum x^2 \end{pmatrix} = \begin{pmatrix} \sum y^2 - (\sum x)^2 & n\sum xy - (\sum x)(\sum y) \\ n\sum xy - (\sum x)(\sum y) & n\sum x^2 - (\sum x)^2 \end{pmatrix}$$

9. ANOVA FOR SLRM (ANALYSIS OF VARIANCE)

• tells us about goodness of fit of model.

$$SSE = \sum e^2 = \sum (y - \hat{y})^2 \rightarrow \text{error}$$

$$\therefore SSE = SST - SSR$$

$$SST = \sum (y - \bar{y})^2 \rightarrow \text{total}$$

$$SSR = \sum (\hat{y} - \bar{y})^2 \rightarrow \text{regression}$$

Source of variation		Sum of square		Mean of square		F-table
SOV	degree of freedom	SS	MS	F		
reg	$K-1$	SSR	$\frac{SSR}{df} = MSR$	$F = \frac{MSR}{MSE}$		
error	$n-2$	SSE	$\frac{SSE}{n-2} = MSE$			
total	$n-1$	SST				

∴ K = number of regressors, independent variables

P = parameters.

* goodness of fit using anova.

1. H_0 : model is not best fit

H_1 : model is best fit

2. $\alpha = 0.05$, etc.

3. $TS \rightarrow F = \frac{MSR}{MSE}$

4. CR

$$F_c > F_{\alpha}(v_1, v_2) \quad \begin{matrix} p-1 \\ n-p \end{matrix}$$

∴ calculated value lies in CR, so reject H_0 .

5. MLRM

independent variables ≥ 2

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + e$$

∴ k independent variables (x)
k+1 parameters

special case k=2

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + e$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

$$\sum y = n\beta_0 + \beta_1 \sum x_1 + \beta_2 \sum x_2$$

$$\sum x_1 y = \beta_0 \sum x_1 + \beta_1 \sum x_1^2 + \beta_2 \sum x_1 x_2$$

$$\sum x_2 y = \beta_0 \sum x_2 + \beta_1 \sum x_1 x_2 + \beta_2 \sum x_2^2$$

$$\begin{pmatrix} n & \sum x_1 & \sum x_2 \\ \sum x_1 & \sum x_1^2 & \sum x_1 x_2 \\ \sum x_2 & \sum x_1 x_2 & \sum x_2^2 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 7 \\ 6 \\ 8 \end{pmatrix}$$

3 independent variables

$$A\beta = b \quad \beta = A^{-1}b$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

$$\sum y = n\beta_0 + \beta_1 \sum x_1 + \beta_2 \sum x_2 + \beta_3 \sum x_3$$

$$\sum x_1 y = \beta_0 \sum x_1 + \beta_1 \sum x_1^2 + \beta_2 \sum x_1 x_2 + \beta_3 \sum x_1 x_3$$

$$\sum x_2 y = \beta_0 \sum x_2 + \beta_1 \sum x_1 x_2 + \beta_2 \sum x_2^2 + \beta_3 \sum x_2 x_3$$

$$\sum x_3 y = \beta_0 \sum x_3 + \beta_1 \sum x_1 x_3 + \beta_2 \sum x_2 x_3 + \beta_3 \sum x_3^2$$

$$\begin{pmatrix} n & \sum x_1 & \sum x_2 & \sum x_3 \\ \sum x_1 & \sum x_1^2 & \sum x_1 x_2 & \sum x_1 x_3 \\ \sum x_2 & \sum x_1 x_2 & \sum x_2^2 & \sum x_2 x_3 \\ \sum x_3 & \sum x_1 x_3 & \sum x_2 x_3 & \sum x_3^2 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}$$

BAYESIAN STATISTICS

Prior + likelihood

Posterior

★ 3 important components.

1. prior information
Uniform bernoulli

2. likelihood

$$L = f(x_1) f(x_2) f(x_3) \dots$$

3. posterior

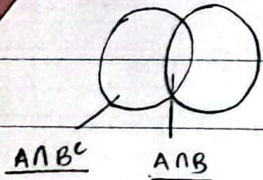
$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

nothing in common

all

A and B forms partition of S, if they're mutually exclusive (ME) and collectively exhaustive.

Law of Total Probability.



$$A = (A \cap B) \cup (A \cap B^c)$$

$$P(A) = P[(A \cap B) \cup (A \cap B^c)]$$

$$P(A) = P(A \cap B) + P(A \cap B^c) - P[(A \cap B) \cap (A \cap B^c)]$$

$$P(A) = P(A|B) \cdot P(B) + P(A|B^c) \cdot P(B^c)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A|B) \cdot P(B)$$

BAYES THEOREM

If S has two partitions $\begin{matrix} B \\ B^c \end{matrix}$; then $\rightarrow P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A|B) \cdot P(B) + P(A|B^c) \cdot P(B^c)}$

$$P(A|B) \cdot P(B) + P(A|B^c) \cdot P(B^c)$$

$$\text{PROOF: LHS} = P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A|B) \cdot P(B)}{P(A|B) \cdot P(B) + P(A|B^c) \cdot P(B^c)}$$

3. JOINT PROBABILITY DISTRIBUTION

in this case we have 2 variables, find JPD and then analyze

1. Marginal properties

2. Independence

3. Conditional Probabilities.

Q. Toss a coin 3 times and record number of heads appeared. let X denote number of heads on 3rd toss.

Y denotes number of heads in 3rd flip.

a) construct table.

b) marginal properties: sum of properties corresponding to 1 value.

c) $P(X < Y)$

d) conditional probabilities: $f(x|y)$ and $f(y|x)$

e) are X and Y independent.

$$S = \{HHH, TTT, HTH, HHT, TTH, THT, THT, HTT\}$$

$$X = \{0, 1\}$$

$$Y = \{0, 1, 2, 3\}$$

a)

		Y				
		0	1	2	3	$g(x)$
X	0	$1/8$	$2/8$	$1/8$	$0/8$	$4/8$
	1	0	$1/8$	$2/8$	$1/8$	$4/8$
		$h(y)$	$1/8$	$3/8$	$3/8$	$1/8$
						1

$$d) f(x|2) = f(0|2) = \frac{f(0,2)}{h(2)} = \frac{1/8}{3/8}$$

$$f(x|2) = f(1|2) = \frac{f(1,2)}{h(2)} = \frac{2/8}{3/8}$$

$$f(1|Y) = f(1|0) = \frac{f(1,0)}{h(0)} = \frac{0}{1/8}$$

$$f(1|Y) = f(1|1) = \frac{f(1,1)}{h(1)} = \frac{1/8}{3/8}$$

$$f(1|Y) = f(1|2) = \frac{f(1,2)}{h(2)} = \frac{2/8}{3/8}$$

$$f(1|Y) = f(1|3) = \frac{f(1,3)}{h(3)} = \frac{1/8}{1/8}$$

b)

X	$g(x)$	Y	$h(y)$
0	$4/8$	0	$1/8$
1	$4/8$	1	$3/8$
		2	$3/8$
		3	$1/8$

c) $P(X < Y)$

$$f(0,1) + f(0,2) + f(0,3) + f(1,2) + f(1,3)$$

$$= \frac{2}{8} + \frac{1}{8} + \frac{0}{8} + \frac{2}{8} + \frac{1}{8} = \frac{6}{8}$$

e) independence

2 variables x and y will be independent if $f(x, y) = g(x) \cdot h(y)$

$$f(0, 0) = g(0) \cdot h(0)$$

$\neq$ is dependent

$$1/8 = \frac{4}{8} \cdot \frac{1}{8}$$

Q. Two marbles are selected containing 3 black, 2 red,

if x denotes number of black marbles

y denotes number of red marbles.

a) construct table

b) marginal property

c) $P(X+Y \leq 1)$

d) $f(x|1)$, $f(2|y)$

e) check independence.

$$X = \{0, 1, 2\}$$

$$Y = \{0, 1, 2\}$$

		Y			
		0	1	2	$g(y)$
X	0				
	1				
	2				
	$h(y)$				